Do Now · retrieval

Lesson 059 · Thinking Abstractly: What's the Model?

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What makes a good field in a record?

Q2

Why give a unique record key?

Q3

What is a precondition?

Q4

Trace binary search for 4 in the sorted, zero-indexed array [2,4,6,8,10]. Start with low=0 and high=4. Use mid=(low+high) DIV 2, high=mid-1 or low=mid+1. State the indexes checked, in order.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Abstraction selects only the information required for the task and deliberately excludes irrelevant real-world detail.

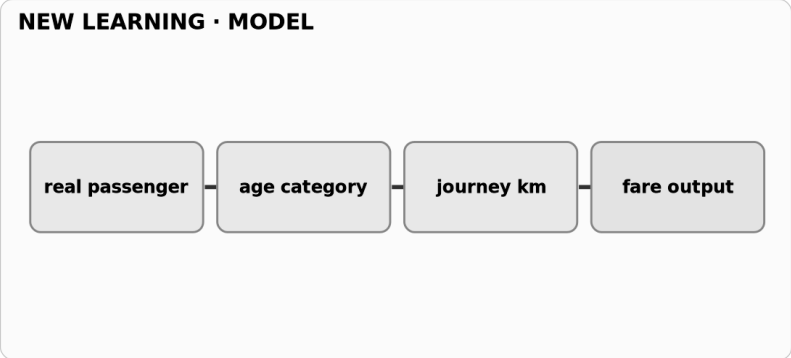
Core idea 2

Model — An abstract model is a simplification of reality: here distance and child/adult status determine the fare; hair colour does not.

Core idea 3

Evaluate — The model is only as good as its assumptions. If peak pricing or concessions apply, the abstraction needs refinement.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

abstraction — Focus on relevant details.

model — Simplified representation of reality.

assumption — Condition accepted within a model.

irrelevant — Detail that does not affect task.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

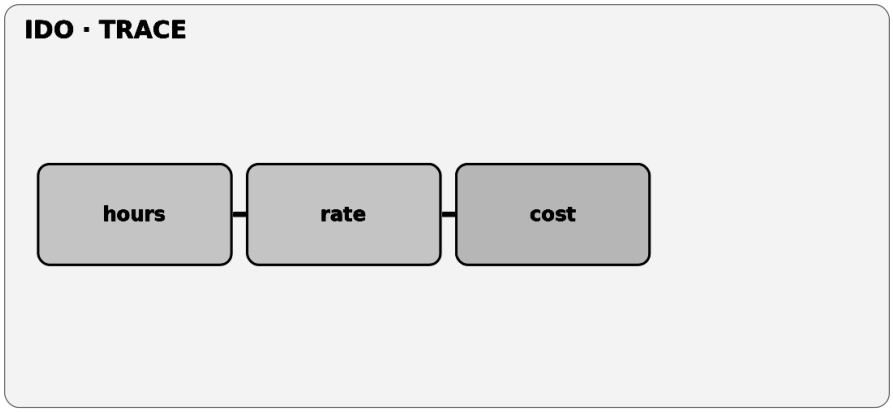
Problem

Choose relevant inputs for calculating parking fee at £2 per hour, rounded to whole hours.

Follow the worked method

1. Explain — ParkingFee(hours)=hours×£2 for the simplified model; define non-negative hours and rounding rules.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
Fee=0.20×lateDays in this simplified rule; real libraries might cap fees or exempt users.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For an automated bus fare model, identify three relevant and two irrelevant details. [5]

Question 2

Define inputs, processing and output for library overdue-fee model at 20p/day capped at £2. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate the risks of abstracting a school attendance intervention down to attendance percentage alone. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An abstraction must include every observable detail of reality.

Use key vocabulary from memory

1. Explain abstraction in your own words.

2. Explain model in your own words.

3. Explain assumption in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.